Yuchen Wang

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Education

Harbin Institute of Technology, Shenzhen Sep 2021 – Jun 2025
BEng in Computer Science

- **Average Score:** 89.818/100

The Hong Kong University of Science and Technology, Guangzhou Sep 2025 – Jun 2027
Mphil in Data Science and Analytics | Supervised by Prof. Zeke Xie

- **CGA:** 4.30/4.30

Research Experience

Handling Imbalanced Pseudolabels for Vision-Language Models Sep 2024 – May 2025
In Proc. of ICML 2025 | First author

- Developed strategies to enhance data utilization efficiency in weakly supervised and unsupervised settings by adapting VLMs to downstream image classification tasks through pseudolabeling.
- Focused on the issue of category imbalance in pseudolabels arising from biased predictions of VLMs.
- Investigated the root causes of imbalanced predictions given by VLMs in representation space, identifying two key factors: concept mismatch and concept confusion.
- Proposed a framework with concept alignment and confusion-aware calibrated margin to improve the utilization of unlabeled data, achieving a 6.29% relative improvement over SOTA methods.

Exploring Data-Free LoRA Transferability for Video Diffusion Models Oct 2025 – May 2026
Accepted to ICML 2026 | First author

- Investigated the transferability of personalized adaptations across video diffusion models, enabling reuse on distilled variants without additional training data.
- Analyzed the weight-space effects of full fine-tuning and LoRA modules, uncovering structured behaviors in singular subspaces (e.g., spectral rigidity and cluster-level routing).
- Identified transfer failure as interference between incompatible routing patterns induced by full fine-tuning and LoRA.
- Proposed a data-free cluster-aware arbitration framework to restore adaptation effects while preserving the target generative space, improving both generation stability and style fidelity.

Publications

Wang, Y., Bai, X., Li, X., Guan, W., Nie, L., Chen, X. Handling Imbalanced Pseudolabels for Vision-Language Models with Concept Alignment and Confusion-Aware Calibrated Margin. ICML 2025. [Paper]

Wang, Y., Zhong, W., Bai, L., Zhou, Z., Shao, S., Cheng, B., Chen, S., Yang, S., Xie, Z. Exploring Data-Free LoRA Transferability for Video Diffusion Models. ICML 2026 (Accepted). [Paper]

Honors and Awards

Outstanding Undergraduate Thesis (top 3%), Harbin Institute of Technology, Shenzhen 2025
Bronze Medal, The 2022 ICPC Asia Hefei Regional Contest 2022
First Prize, Lanqiao Cup Guangdong Provincial Contest 2022,2023
Undergraduate Academic Scholarship, Harbin Institute of Technology 2022, 2023